SOFTWARE REQUIREMENTS SPECIFICATION

for

CareBridge

A Structured Nurse Handoff and Patient Status Management System

Version 1.0

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October 15, 2026

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1 Introduction

1.1 Purpose

A nursing shift handoff is when one nurse passes responsibility for a patient to another. It happens several times a day on every ward, and what the incoming nurse knows depends on what the outgoing nurse remembers to say. Today this is verbal and leaves no record.
"CareBridge" fixes this. It gives nurses a structured place to log patient status during the shift, and a way to hand that record to the next nurse with accountability. This document specifies the requirements of CareBridge version 1.0.

1.2 Intended Audience and Reading Suggestions

This SRS is for developers, project managers, users (nurses, doctors, administrators), testers and evaluators. Developers and testers should focus on Chapters 3 and 4. Managers and evaluators can start with Chapters 1 and 2.

1.3 Project Scope

"CareBridge" lets Nurses, Doctors and Administrators manage patient status and shift handoffs on a hospital ward.
A nurse signs in and sees her assigned patients. During the shift she adds notes to each patient under **Details** and **Major Decisions**. At the end of the shift she picks the patients to hand over, edits the text, and sends each handoff to a named nurse.
The incoming nurse reads the record and responds with **Accept**, **Decline** or **Report Issue**. A reason is required for Decline and Report Issue. Every step is saved in an append-only audit trail, and Time-to-Acknowledgement (TTA) comes directly from stored timestamps.
Doctors view logs of their assigned patients and receive nurse alerts. Administrators manage nurse accounts and assign nurses and doctors to patients.
For this release the **nurse workflow is built end to end**. The doctor and administrator interfaces are deferred (Section 3.2.6), but the schema already supports them.

Figure 1.1 gives an overview of the system.
Outgoing nurse: views assigned patients, updates status, reviews details and major decisions, and submits the handoff.
Incoming nurse: views the handoff, reviews the patient log, and accepts, declines or

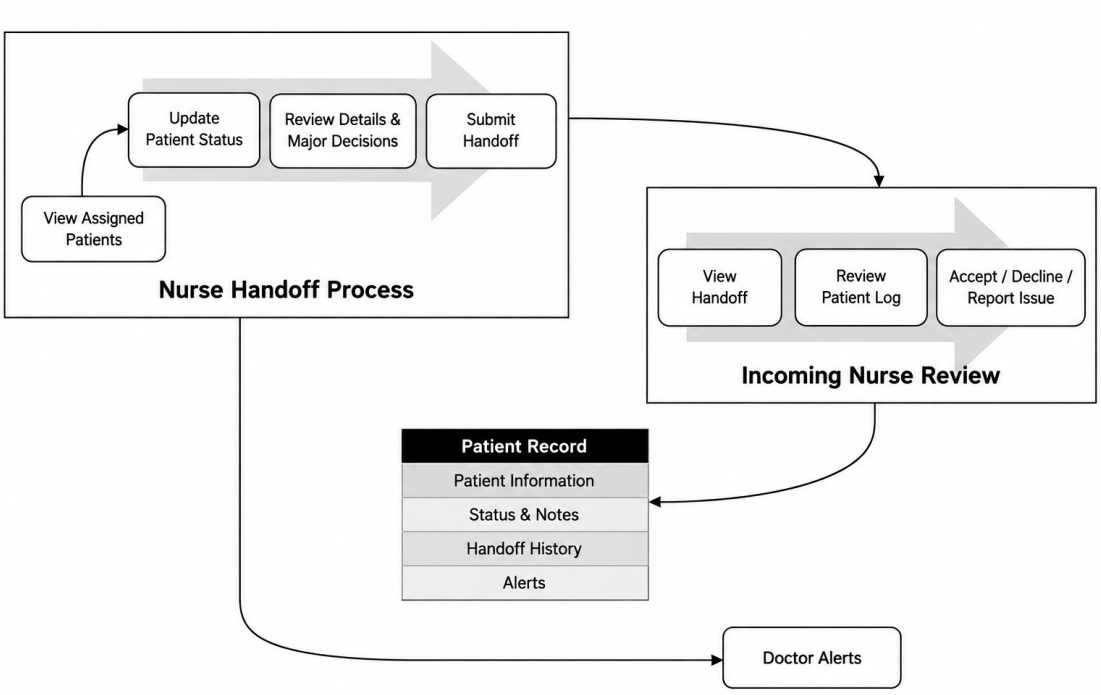


Figure 1.1: Entire work-flow

reports an issue. The result updates the Patient Record, and escalations reach the doctor as Doctor Alerts.

1.4 Definitions and Abbreviations

Term	Meaning
SRS	Software Requirements Specification
TTA	Time-to-Acknowledgement: time between a handoff being sent and answered
Handoff	Transfer of one patient from an outgoing to an incoming nurse
RLS	Row Level Security (PostgreSQL)
JWT	JSON Web Token issued by Supabase Auth

2 Overall Description

2.1 Product Perspective

"CareBridge" replaces the verbal, memory-based handoff. Five problems were seen on the ward:

1. Nothing is written down consistently.
2. Handoff quality depends on who gives it and how tired they are.
3. There is no record of who said what, or if it was acknowledged.
4. There is no formal way to push back on an unclear or misaddressed handoff.
5. Nothing can be measured.

All five come from one cause: the handoff produces no record. CareBridge makes the handoff a real record and builds accountability on it.

2.1.1 System Architecture

CareBridge has three tiers on top of a managed database.

- **Client:** a React single-page app in a Tauri shell, with no business logic.
- **Application:** a FastAPI service that owns every rule and is the only component that talks to the database.
- **Data:** Supabase PostgreSQL, with Row Level Security as a second layer under the application's checks.

The client never talks to the database directly. This keeps the rules in one place and the database secret key off client devices.

2.2 User Classes and Characteristics

"CareBridge" has 3 types of users.

- Nurses
  - Outgoing Nurse
  - Incoming Nurse

- Doctors
- Administrators

A nurse is outgoing when handing over her patients and incoming when receiving a handoff.

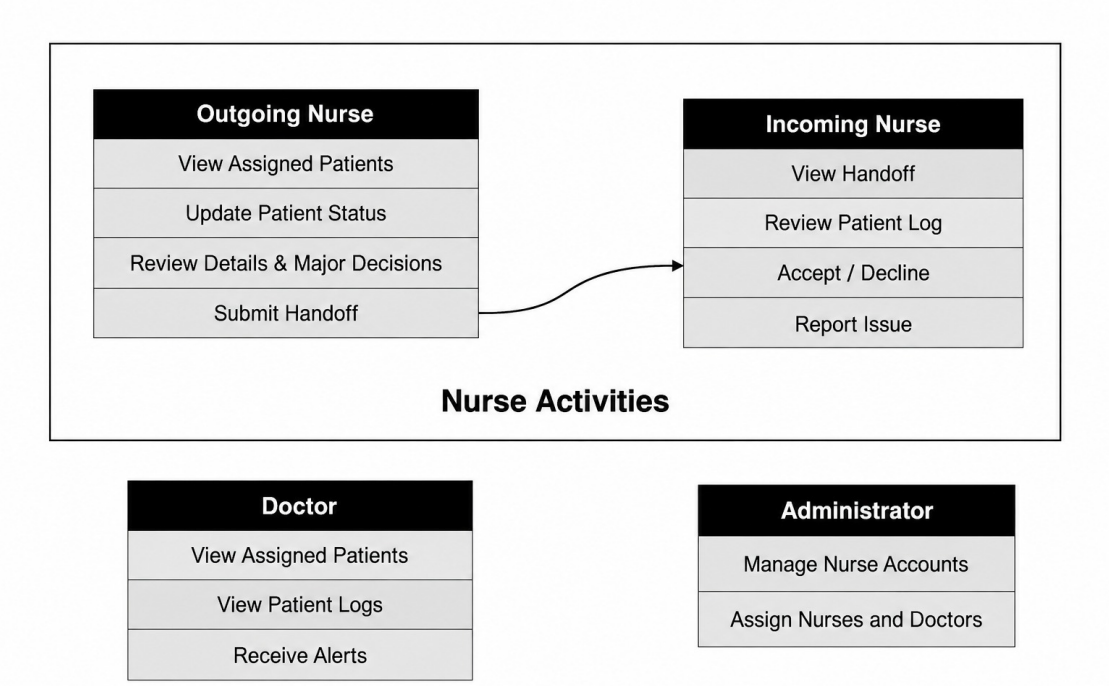


Figure 2.1: Type of users and their activities

Figure 2.1 shows what each user can do: the Outgoing Nurse (view patients, update status, review details, submit handoff), the Incoming Nurse (view handoff, review log, accept or decline, report issue), the Doctor (view patients and logs, receive alerts) and the Administrator (manage nurse accounts, assign nurses and doctors).

2.3 Product Functions

Figure 2.2 shows how the entities interact. The Administrator assigns both nurses. The Outgoing Nurse updates the Patient and creates a Handoff, which is stored with the Patient and in the Database and sent to the Incoming Nurse. The Incoming Nurse reviews the Patient and alerts the Doctor, and the Doctor views the Patient.
Users must sign in first. Identity comes only from the Supabase Auth JWT, and no endpoint accepts a nurse ID in the request body.
Staff have - id, full_name, email, role (nurse, doctor, admin), active, created_at.

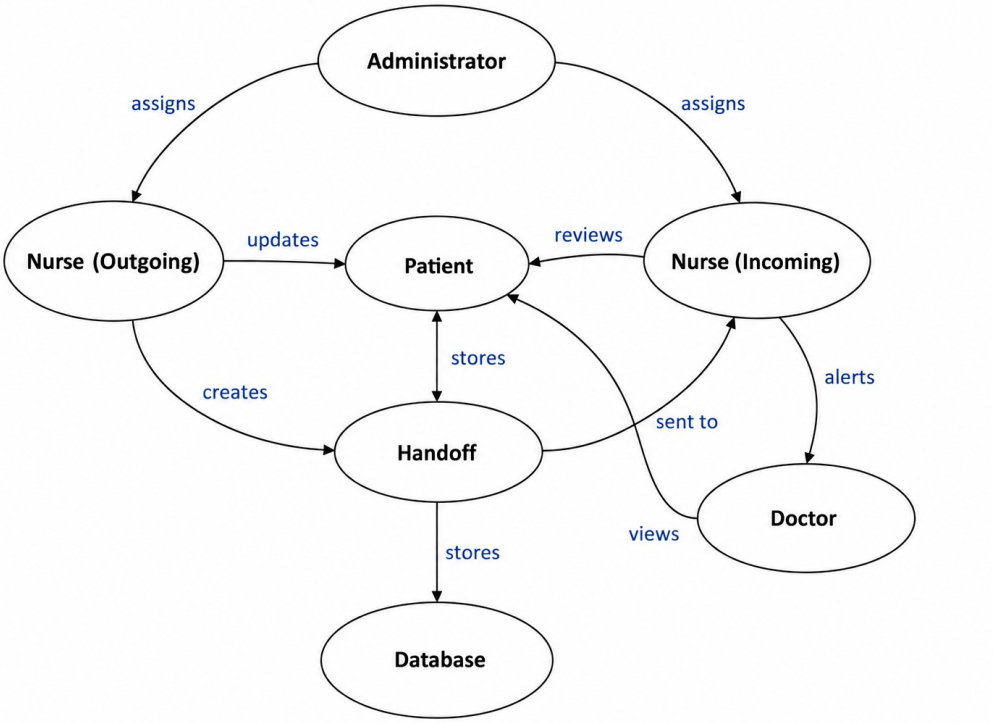


Figure 2.2: Entity interaction diagram

Patients have - patient_id, name, date_of_birth, gender, room_number, status, assigned_nurse_id, assigned_doctor_id, admitted_at.
Shifts have - nurse_id, started_at, ended_at, status.
Patient notes have - patient_id, author_id, shift_id, section (detail or major decision), body, created_at. Notes are append-only.
Handoffs have one row per patient - patient_id, from_nurse_id, to_nurse_id, details, major_decisions, status, response_reason, sent_at, responded_at.
Handoff events are the audit trail - handoff_id, actor_id, event (sent, viewed, accepted, declined, issue reported), created_at.
Alerts are nurse-to-doctor escalations - patient_id, raised_by, doctor_id, message, status, created_at.

2.4 Operating Environment

The client runs in a Tauri v2 shell on Windows, macOS or Linux, and its layout suits shared ward tablets. The backend uses Python 3.13 with FastAPI, and the database is Supabase (PostgreSQL 17).

2.5 Design

A nurse handoff has 3 steps -

- Prepare: notes for each selected patient pre-fill the Details and Major Decisions boxes. Nothing is saved yet.
- Review: the nurse edits the text and picks a named recipient. Sending creates one handoff per patient and logs a *sent* event.
- Respond: the receiving nurse opens the handoff (logs a *viewed* event), reads the record and responds.

The three responses are:

- **Accept** transfers the patient to her.
- **Decline** needs a reason and leaves the patient with the sender.
- **Report Issue** needs a reason and still transfers the patient, so coverage never lapses while the problem is sorted out.

Each response sets the responded time and adds an event. It only applies if the handoff is still pending; a second response gets HTTP 409.

2.6 Constraints and Assumptions

- Only the nurse workflow is built in this release.
- All rules are enforced by FastAPI, and notes and audit events are append-only.
- Until the admin interface exists, patients, accounts and assignments are created directly in the database.
- The system depends on Supabase Auth and PostgreSQL being available.

3 System Features

"CareBridge" is a structured handoff system, so its main work is recording notes, sending handoffs and capturing responses.

3.1 Description and Priority

Features in priority order -

1. Per-patient Handoff : the main feature. A nurse picks patients and sends each handoff to a named nurse.
2. Handoff Response : Accept, Decline or Report Issue, with a reason for the last two.
3. Patient Record : Details, Major Decisions and a chronological Log.
4. Identity and Shift Lifecycle : sign-in, roles, opening and closing shifts.
5. Audit Trail and TTA : append-only events and timestamps.
6. Access Control : a nurse reaches only her own patients or one being handed to her.
7. Doctor Alerts and Administration : deferred.

3.2 Functional Requirements

3.2.1 Identity and Shift Lifecycle

- FR-1: The system shall authenticate users with Supabase Auth and resolve the JWT to a staff record.
- FR-2: No request shall accept a user or nurse ID in its body.
- FR-3: Requesting the current shift shall return the open shift or open a new one, so refreshing never creates duplicates.
- FR-4: Closing a shift with patients still assigned and no handoff sent shall be rejected (HTTP 409).
- FR-5: A nurse shall have only one active shift.

3.2.2 Patient Record

- FR-6: A nurse shall view her assigned patients.
- FR-7: A nurse shall add notes to her own patients under Details or Major Decisions.
- FR-8: Each patient screen shall show Details, Major Decisions and a full chronological Log.
- FR-9: Notes shall be append-only, with no update or delete path.

3.2.3 Per-Patient Handoff

- FR-10: A nurse shall select one or more of her patients to hand over.
- FR-11: The system shall pre-fill the text from the shift's notes without saving anything.
- FR-12: The nurse shall edit the text and choose a named recipient.
- FR-13: Sending shall create one handoff per patient and log a *sent* event.
- FR-14: A handoff shall not be sent to oneself.
- FR-15: Each patient shall have at most one pending handoff.

3.2.4 Handoff Response

- FR-16: The receiving nurse shall be able to read the record before responding, and opening the handoff shall log a *viewed* event.
- FR-17: She shall respond with Accept, Decline or Report Issue.
- FR-18: Accept shall transfer the patient to her.
- FR-19: Decline shall need a non-empty reason and leave the patient with the sender.
- FR-20: Report Issue shall need a non-empty reason and still transfer the patient.
- FR-21: Every response shall set the responded time and add an event.
- FR-22: A response shall apply only to a pending handoff, otherwise HTTP 409.
- FR-23: A sender shall not respond to her own handoff.

3.2.5 Audit and Measurement

- FR-24: The system shall record sent, viewed, accepted, declined and issue-reported events with actor and time.
- FR-25: A resolved handoff shall always have a responded time, and a pending one never.
- FR-26: TTA and the clean-accept rate shall be computable from stored data, for example:

```
SELECT avg(responded_at - sent_at)
FROM handoffs
WHERE responded_at IS NOT NULL;
```

3.2.6 Doctor and Administrator (Deferred)

- FR-27: A doctor shall view assigned patients and logs, and receive nurse alerts.
- FR-28: An administrator shall manage nurse accounts and assign nurses and doctors.
- FR-29: Voice notes (with mandatory review) and push notifications.

These are out of scope for version 1.0. The alerts table and doctor column already exist in the schema.

3.3 Data Requirements

Figure 3.1 shows the entity-relationship diagram. Notes live in one table with the section as a column, so the sectioned views and the log always match. A handoff belongs to a patient, not a whole shift, so one patient can be declined without rejecting the batch.

3.3.1 Rules Enforced by the Database

- A handoff cannot be sent to yourself.
- Decline and Report Issue need a non-empty reason.
- A resolved handoff has a responded time, a pending one does not.
- One pending handoff per patient and one active shift per nurse.

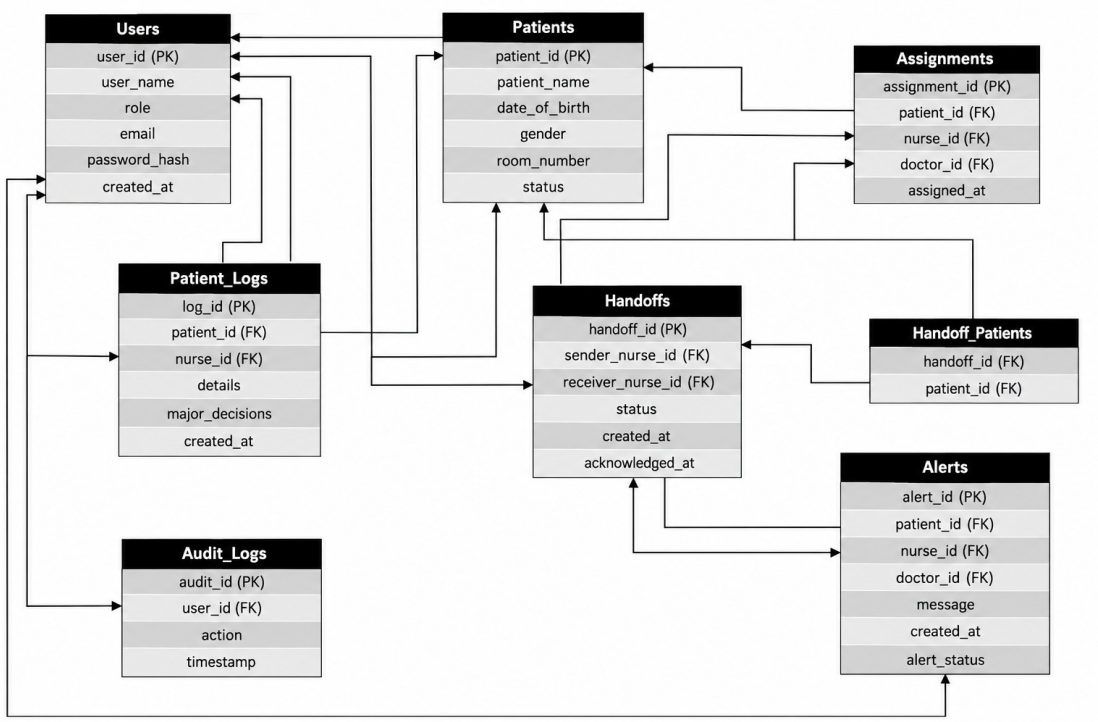


Figure 3.1: Entity-Relationship Diagram

3.3.2 Technical Stack

- Front-End : React 19, TypeScript, Vite, React Router, Tailwind CSS, Tauri v2.
- Back-End : Python 3.13, FastAPI, Uvicorn, Pydantic v2.
- Database and Auth : Supabase (PostgreSQL 17) with RLS.
- Testing : pytest with FastAPI's TestClient.

3.4 External Interface Requirements

3.4.1 User Interfaces

There are three main screens - Shift, My Patients and Handoff - plus screens for a single patient, a note composer and handoff review. All except login sit behind a RequireAuth guard. Colour marks states only: amber for pending, emerald for accepted, rose for declined.

3.4.2 Software Interfaces

The client calls the FastAPI backend over HTTP with a bearer token. The backend has 16 endpoints across five routers (staff, shifts, patients, notes, handoff) and uses Supabase PostgreSQL and Auth.

3.4.3 Communication Interfaces

All communication uses HTTPS, with CORS limited to the deployed client's origin.

4 Other Nonfunctional Requirements

4.1 Performance Requirements

Each screen sends several requests at once, so the backend uses the asynchronous database client. Verified tokens are cached briefly to avoid a round trip to the auth server on every request.

4.2 Security Requirements

- Only signed-in users can access the system; anonymous access is revoked.
- Identity comes only from the verified JWT.
- Access is checked in the application (services/access.py, on every route) and by RLS on all seven tables.
- Viewing a patient (`ensure_can_view_patient`, which includes one being handed to her) is kept separate from editing one (`ensure_assigned_patient`).
- The frontend holds only the publishable key. The backend refuses to start if it is given one.
- Every state change is recorded with the actor.

4.3 Reliability and Data Integrity

Key rules are database constraints. Handoff responses use a conditional update so two responses cannot overwrite each other.

4.4 Software Quality Attributes

Testing is done at four levels:

- Route tests: pytest checks the handoff state machine and note recording, including the actual writes made.
- Constraint tests: each database rule is tested by trying the forbidden operation in a rolled-back transaction.

- Integration checks: schema and seed data are checked on the live Supabase instance.
- Build and security checks: frontend type-check, and the Supabase security advisor after each migration.

4.5 Business Rules

"CareBridge" exists to make handoffs addressed, answerable, auditable and measurable. Patient coverage must never lapse while a documentation problem is sorted out.

4.6 Measurability

The main metric is TTA, taken from the sent and responded times. No measured values are claimed yet, as that needs a pilot on a real ward.

5 Other Requirements

"CareBridge" needs ongoing maintenance. Future work: doctor and administrator interfaces, a pilot with 10–15 nurses (with a baseline measured first), voice notes with mandatory review, backend deployment, push notifications, and AI suggestions for summaries and note categories that the nurse always confirms or edits.